

Short Response: Quantitative Political Risk

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1. Political Risk Use Case

In April 2023, two of Sudan's most powerful military factions, the Sudanese Armed Forces (SAF) and the Rapid Support Forces (RSF), began fighting after months of political tension. The fighting has spread across the country and shows no sign of stopping. This project asks a specific question: does armed conflict near gold mines in Sudan cause that mine to produce less gold? And if so, can we measure that relationship quantitatively to give mining companies and investors an early warning signal?¹

a. Type of Political Risk

This is a security risk, specifically, an intrastate armed conflict where fighting between two armed groups disrupts economic activity on the ground. It also has a geopolitical dimension, since outside actors including Gulf states have supported different sides. This has turned a domestic power struggle into something more complicated.

b. Commercial and Market Impact

Sudan is one of Africa's largest gold producers.² When fighting breaks out near a mine, workers flee, supply trucks stop running, and equipment gets abandoned or looted. The commercial problem is straightforward: a mining company operating in Sudan, or an investor holding stock in one, needs to know whether a specific mine is at risk of shutting down. It is not just about whether Sudan is overall a dangerous country. A precise risk signal, tied to conflict near gold mines, would help companies decide when to pull staff out, when to activate insurance clauses, and whether to continue investing.

c. Why This Is a Political Risk Now

The RSF started as a government-backed militia that was given control over artisanal gold mining in Darfur as a way of financing its operations. When the RSF and the formal military (SAF) turned on each other in April 2023, they brought that fight directly into Sudan's gold-producing regions. This is an active war with a clear start date, identifiable actors, and a direct economic footprint in the mining sector. The conflict has already displaced over 10 million people and effectively halted formal economic activity in large parts of the country.

d. Client or Customer Affected

The main client is a mining company with operations in Sudan, or a commodity-focused investor with exposure to Sudanese gold output. A secondary client could be a political risk insurer trying

¹Sundberg, Ralph and Erik Melander (2013). "Introducing the UCDP Georeferenced Event Dataset." *Journal of Peace Research* 50(4): 523–532.

²World Gold Council, "Gold Production by Country," 2023. Sudan ranked among Africa's top gold producers prior to the 2023 conflict outbreak.

to price coverage for Sudan assets, or a bank deciding whether to extend financing to a mining project there.

2. Current Approaches and Limits

a. Conventional Methodology

The standard way to handle this kind of risk is to hire a political risk consulting firm, like the Economist Intelligence Unit. These types of firms publish a country-level risk score for Sudan.³ Sudan is labeled as “extreme risk,” and companies are advised to avoid or minimize exposure, and analysts write narrative reports describing what might happen. On the ground, companies use private security consultants and incident monitoring services to track the conflict in real time.

b. Limitations

The problem with this approach is that it treats all of Sudan as equally dangerous.⁴ A mine in the Red Sea Hills, far from the fighting, gets the same risk rating as a mine in North Darfur, right in the middle of it. That is not useful for a company trying to decide which specific assets to protect or exit. Country-level scores also update slowly and rely heavily on expert judgment, which is prone to bias and hard to act on quantitatively. There is currently no systematic, data-driven way to say: “this mine’s output is X% likely to drop next month based on current conflict patterns.”

3. Pre-Research Design

a. Quantitative Measurement

Our approach is geography based. For each known mine in Sudan, we will calculate how much conflict happened nearby each month, specifically, the total number of fatalities from all recorded conflict events within 50 km of that mine. This gives a “conflict intensity” score for each mine, each month. We can then test whether months with high nearby conflict intensity are followed by drops in gold output one to two months later.⁵

To keep the analysis clean, we will filter the conflict data to only include events from the SAF-RSF war specifically, and only use events where the location is known with reasonable precision, the dataset includes a precision score, and we will exclude events where the location is only known at the country level.

b. Dataset

The main dataset is the UCDP Georeferenced Event Dataset (GED), version 25.1. This is a publicly available dataset that records individual violent events around the world from 1989 to 2024, with GPS coordinates precise down to the village level, exact dates, and fatality counts. It is a good fit because it covers the Sudan conflict in detail, the location data is precise enough to

³Bremmer, Ian and Preston Keat, *The Fat Tail: The Power of Political Knowledge in an Uncertain World*. Oxford University Press, 2009.

⁴GeoQuant Methodology Overview (course reading); see also *Managing Risk in an Unstable World*, Harvard Business Review, 2005.

⁵Buhaug, Halvard and Jan Ketil Rød. "Local Determinants of African Civil Wars." *Political Geography* 25(3), 2006.

calculate distances to mines, and it is free to download. Mine locations will come from the USGS Mineral Resources Data System or S&P Global's mining database, both of which provide geocoded records of mine sites. Gold export volumes from the World Bank World Development Indicators will serve as the outcome variable.

c. Pre-Registered Hypothesis

Before looking at the data, we expect that higher conflict intensity near a mine, measured by fatalities within 50 km in a given month, will predict lower gold output in the following one to two months. We think this effect will be strongest in Darfur and Kordofan, where the RSF has been most active and where artisanal gold mining is concentrated. We expect weaker effects in the Red Sea Hills, which has seen less fighting. The underlying logic is when armed groups are fighting near a mine, it becomes too dangerous to keep working there.